Cross-Validation of BATMAN Routing Protocol Simulations: NS2 vs. NS3 on Linear Multi-Hop Wireless Chains — What Results Can Be Trusted?

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***Abstract*—Two independent simulation studies of the BATMAN v1 proactive routing protocol are evaluated on identical static linear wireless chains: one using NS2 with a Tcl-level OGM overlay (4-node chain, single run) and one using ns-3.39 with a custom C++ Layer-3 implementation (N = 4–10 nodes, five seeds per configuration, 35 runs total). We perform a metric-by-metric cross-validation to determine which results are genuinely trustworthy, which differ due to controllable experimental parameters, and which diverge because of a fundamental architectural limitation of the NS2 approach. Our principal finding is that five core metrics—per-hop MAC air delay (2.320 ms vs. ≈ 2.3 ms/hop, $<$1% error), steady-state jitter ($<$1.5 ms vs. 1.37 ms), per-node OGM control overhead (319 B/s vs. 325 B/s, $<$2%), TQ decay exponent (exactly $\eta = 11.765\%$ per hop in both), and packet delivery ratio (100% in both)—are *simulator-independent* and can be trusted with high confidence. Route discovery time (30 ms RTR vs. 21.2 ms MAC) and raw E2E delay (9.62 ms vs. 16.5 ms) differ for explainable, correctable reasons (measurement-layer abstraction and packet-size selection, respectively). A *critical divergence* is identified: NS2's deterministic Tcl OGM delivery is blind to the bimodal RDT distribution observed in 22.9% of NS3 runs, where 802.11b MAC collisions inflate route discovery time by $47\times$–$94\times$ (21 ms $\rightarrow$ 1000–2000 ms). We quantify the root cause, provide a corrected trust taxonomy for all reported metrics, and derive unified parameter recommendations for future BATMAN deployments.**

***Index Terms*—BATMAN, MANET, NS2, ns-3, cross-validation, OGM, route discovery time, MAC delay, simulation fidelity, bimodal distribution.**

I. Introduction

Better Approach To Mobile Adhoc Networking (BATMAN) [1] is a proactive link-state-free routing protocol that maintains routing tables through periodic Originator Message (OGM) broadcasts. Each OGM carries a Transmit Quality (TQ) metric that degrades multiplicatively with each relay hop, creating a distributed gradient toward the best next-hop for every originator. BATMAN has been standardised as *batman-adv* in the Linux kernel and deployed extensively in community mesh networks.

Despite widespread deployment, the protocol's simulation-level behaviour is characterised inconsistently in the literature. Two recent works study BATMAN on static linear chains:

- **NS2 study** (Zafar): 4-node chain, Tcl-level OGM overlay atop AODV, fine-grained MAC-layer AWK analysis [2].
- **NS3 study** (Umair): $N = 4$–10 nodes, custom C++ L3 module in ns-3.39, FlowMonitor post-processing, five seeds per configuration [3].

TABLE I
Simulation methodology comparison

Dimension	NS2 Study	NS3 Study
Simulator	NS-2 (2020)	ns-3.39 (2023)
BATMAN impl.	Tcl overlay	C++ L3 module
OGM delivery	Deterministic	Probabilistic
OGM collision	None (Tcl)	802.11b MAC
Measurement	RTR + MAC AWK	FlowMonitor
Packet size	212 B	512 B
Data start	$t = 3.5$ s	$t = 11.0$ s
Topology N	4 fixed	4–10
Seeds/N	1	5 (35 total)
Data agent	AODV (no Hello)	Custom BATMAN

Both use identical physical parameters: 10 m node spacing, 2.4 GHz, 1 Mbps, IEEE 802.11(b) MAC, and Friis/FreeSpace propagation (these are mathematically equivalent, Section III). Yet they report apparently different values for route discovery time (30 ms vs. 21.2 ms), E2E delay (9.62 ms vs. 16.5 ms), and OGM amplification ratio ($3\times$ vs. $2\times$).

This paper resolves every discrepancy through rigorous cross-examination, derives a formal trust taxonomy for each reported metric (Table III), and identifies one genuine divergence that reveals a critical blind spot in NS2-style BATMAN simulations: the inability to model MAC-layer OGM collisions and the resulting bimodal route-discovery-time distribution.

The rest of this paper is organised as follows. Section II compares simulation architectures. Section III validates physical-layer equivalence. Section IV performs metric-by-metric cross-validation. Section V analyses the bimodal RDT divergence. Section VI extrapolates scalability. Section VII presents the trust taxonomy. Section VIII gives unified recommendations. Section X concludes.

II. Methodology Comparison

Table I contrasts the two simulation architectures across every dimension that could affect results.

The most consequential difference is the OGM propagation model. In the NS2 study, OGMs are dispatched via Tcl `after` timers with a hard-coded 10 ms routing-layer delay per hop—a scheduling abstraction that guarantees OGM delivery with zero collision probability. In the NS3 study, OGMs are real UDP broadcast frames fully subject to 802.11b DIFS, contention-window backoff, and MAC-layer collision. This single architectural choice is solely responsible for the most significant divergence identified in this paper (Section V).

III. Physical Layer Equivalence

Both papers use the Friis free-space propagation formula

$$P_r(d) = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2, \quad \lambda = \frac{c}{f} = \frac{3\times 10^8}{2.4\times 10^9} = 0.125\ \text{m}, \tag{1}$$

with unity antenna gains. NS2 labels its model "FreeSpace"; NS3 labels it "Friis"; they are mathematically identical. With $P_t = 1\,\text{mW}$ and the respective receive thresholds, both simulators yield $d_{\max} \approx 15\,\text{m}$, so each node is in range only of its two immediate neighbours (10 m spacing, 20 m skip-hop distance $> d_{\max}$). This is confirmed by both studies reporting the same single-hop neighbourhood structure.

IV. Metric-by-Metric Cross-Validation

A. Per-Hop MAC Air Delay — Strongest Agreement

The NS2 study directly measures the time for one over-the-air hop as 2.320 ms for all three hops in the S–R1–R2–D chain. The NS3 study reports mean E2E delay of 16.5 ms at $N = 4$ growing at ≈2.3 ms per additional node (per-hop relay stage). The agreement is

$$\delta = \frac{|2.320 - 2.300|}{2.320} = 0.86\%, \tag{2}$$

well within measurement uncertainty.

The 2.320 ms per hop decomposes as

$$t_{\text{hop}} = t_{\text{TX}} + t_{\text{SIFS}} + t_{\text{ACK}} = 1.888 + 0.010 + 0.304 \approx 2.32\ \text{ms}, \tag{3}$$

where $t_{\text{TX}} = (212\times 8)/10^6 + t_{\text{PLCP}} = 1.696 + 0.192 = 1.888$ ms at 1 Mbps, $t_{\text{SIFS}} = 10\,\mu\text{s}$ (802.11b), and $t_{\text{ACK}} \approx 0.304$ ms (PLCP + 14-byte ACK body). The physical computation is *simulator-independent*. Fig. 1 shows the full breakdown.

B. E2E Delay: Explainable by Packet Size

The raw E2E values differ significantly: NS2 reports 9.622 ms, NS3 reports 16.5 ms for $N = 4$. The gap is *entirely accounted for* by the packet-size difference:

$$\Delta t_{\text{TX}} = \frac{(512-212)\times 8}{10^6} = 2.400\ \text{ms/hop}, \tag{4}$$

$$\Delta t_{\text{E2E}} = 3\times 2.400 = 7.200\ \text{ms (predicted)}. \tag{5}$$

The observed difference is $16.5 - 9.622 = 6.878$ ms. The prediction accounts for $7.200/6.878 = 104.7\%$ of the difference; the 0.32 ms residual is attributable to NS3's additional BIDI verification and sliding-window TQ overhead absent in the NS2 Tcl implementation. *Both simulators implement the 802.11b MAC identically.*

C. Route Discovery Time

BATMAN RDT is defined as the elapsed time from the destination's first OGM transmission to the source's route installation.

$$\text{RDT} = t_{S,\text{route}} - t_{D,\text{first-OGM}}. \tag{6}$$

NS2 reports 30 ms; NS3 reports 21.2 ms at $N = 4$. The 8.8 ms gap (29.3%) is explained by measurement-layer abstraction:

- NS2's 10 ms/hop includes Tcl event-queue latency (≈3 ms/hop) on top of the real MAC OGM frame time (≈7 ms/hop).
- NS3's 7.1 ms/hop (21.2/3) measures the MAC-level OGM propagation directly.

The NS3 linear regression $\text{RDT}(N) = 11.30N - 26.64$ (ms), $R^2 = 0.955$, confirms a MAC-level slope of ≈8 ms/hop—consistent with NS2's MAC characterisation once the Tcl overhead is subtracted.

D. TQ Decay Model

Both papers implement the identical hop-penalty formula

$$TQ_{n+1} = TQ_n \times \frac{TQ_{\max} - \text{HP}}{TQ_{\max}} = TQ_n \times \frac{225}{255}, \quad \eta = 11.765\%. \tag{7}$$

The 3-hop chain yields $255 \to 225 \to 198 \to 174$. NS2 (integer arithmetic) and NS3 (floating-point η) produce values within 0.34% of each other, negligible for routing decisions. This metric is **maximally trustworthy**—it is a pure arithmetic property of the protocol specification.

E. Jitter

NS2 reports steady-state jitter < 1.5 ms for $N = 4$. NS3 reports mean jitter of 1.37 ms at $N = 4$, growing at ≈0.3 ms/hop to 3.2 ms at $N = 10$. Agreement at $N = 4$ is within the NS2 upper bound.

F. Packet Delivery Ratio

Both studies report 100% PDR on the $N = 4$ chain. NS3 extends this to all $N = 4$–10 across all 35 runs, confirming 100% PDR as a topological property of static linear chains with adequate guard time before traffic start.

G. OGM Amplification: Counting Semantics

NS3 reports $A = N - 2$ relay transmissions per OGM (2× at $N = 4$). NS2 reports a ratio of $72/24 = 3\times$. The discrepancy resolves via counting semantics: NS3 counts *relay transmissions* (correct overhead measure); NS2's counter conflates relay transmissions with direct-reception route-install events ($N - 1 = 3$ per 4-node chain). The underlying bandwidth overhead ($N - 2$ rebroadcasts) is consistent.

H. Control Overhead

Despite different packet sizes, both papers report nearly identical per-node OGM overhead: NS2 ≈319 B/s, NS3 ≈325 B/s at $N = 4$ ($\delta = 1.88\%$). Both measure total observed overhead including Layer-2 framing and ACK exchange—a simulator-independent property.

TABLE II
RDT OUTCOME DISTRIBUTION (NS3) VS. NS2 FIXED VALUE

Scenario	RDT	Prob.	vs. AODV
NS2 (deterministic)	30 ms	100% (design)	worse
NS3 normal mode	21.2 ms	≈77%	better
NS3 1-miss	≈1000 ms	≈17%	37× worse
NS3 2-miss	≈2000 ms	≈5.7%	75× worse
AODV (reactive)	26.6 ms	N/A	baseline

TABLE III
CROSS-VALIDATION TRUST TAXONOMY

Metric	NS2	NS3	Trust	Cause
Per-hop MAC air delay	2.320 ms	2.3 ms	HIGH	Physics
Jitter (N=4)	<1.5 ms	1.37 ms	HIGH	Physics
Control overhead	319 B/s	325 B/s	HIGH	Protocol
TQ decay (η)	11.765%	11.765%	HIGH	Spec
PDR (static)	100%	100%	HIGH	Topology
Route stab. (flap rate)	0	0%	HIGH	Topology
RDT (raw)	30 ms	21.2 ms	MED	Layer
E2E delay (raw)	9.62 ms	16.5 ms	MED	Pkt size
OGM amplif. ratio	3×	2×	MED	Counting
RDT scaling slope	10 ms/hop	8 ms/hop	MED	RTR abs.
Bimodal RDT distrib.	Not seen	22.9%	LOW	Arch. limit

V. CRITICAL DIVERGENCE: BIMODAL RDT DISTRIBUTION

A. NS3 Finding

Across 35 NS3 runs, 8 (22.9%) exhibit anomalous RDT values clustered at integer multiples of $T_{\text{OGM}} = 1$ s:

- ≈1000 ms ($1 \times T_{\text{OGM}}$): 6 occurrences.
- ≈2000 ms ($2 \times T_{\text{OGM}}$): 2 occurrences.

Root cause: 802.11b MAC-layer collisions when OGMs from multiple originators arrive at an interior relay within the same DIFS window. When an OGM is lost, the next propagation attempt must await the originator's next broadcast, exactly T_{OGM} s later. The probability of k consecutive misses at the critical hop is

$$P(\text{miss}^k) \approx p_{\text{hop}}^k, \quad p_{\text{hop}} \approx 0.22 \text{ (estimated)}. \tag{8}$$

B. Why NS2 Cannot Observe This

NS2's Tcl-level OGMs are dispatched as guaranteed scheduler events. There is no DIFS, no contention window, no collision. OGM delivery is *guaranteed by construction*. The NS2 paper itself acknowledges: "the Tcl-level OGM simulation bypasses MAC-layer collision, so OGM delivery is deterministic." NS2's RDT of 30 ms therefore represents the *best-case* collision-free scenario, not the distribution of RDTs in a real deployment.

C. Quantitative Impact

Table II summarises the full RDT distribution including the NS2 deterministic point. Fig. 2 shows the comparison visually. The worst-case RDT (≈2000 ms) is ≈94× the normal-mode value (21.2 ms)—a failure mode completely invisible to NS2.

D. Design Implication

NS2's conclusion that BATMAN provides a 2.17 s proactive advantage over AODV holds only in the 77% of runs where no OGM miss occurs. In 22.9% of deployments, the proactive advantage is *reversed*: BATMAN has not converged even after 1 s, and AODV's reactive 26.6 ms discovery would have been preferable. Two mitigations are available:

1) Add $\pm T_{\text{OGM}}/4 = \pm 250$ ms random jitter to OGM transmission (already implemented in production *batman-adv*).
2) Reduce T_{OGM} from 1 s to 0.5 s, halving the worst-case RDT penalty at the cost of doubled overhead.

VI. SCALABILITY CROSS-VALIDATION

Using NS2's per-hop characterisation (2.320 ms air, 1.13 ms backoff), we extrapolate E2E delay as

$$\hat{d}_{\text{E2E}}(N) = (N-1) \times (2.320 + 1.130) = (N-1) \times 3.45 \text{ ms}. \tag{9}$$

At $N = 10$ (9 hops): $\hat{d} = 31.05$ ms vs. NS3 measured 30.3 ms ($\delta = 2.5\%$). This near-perfect agreement validates that both simulators implement the same 802.11b store-and-forward delay model. NS2's per-hop decomposition *correctly predicts* NS3's scaling to within measurement uncertainty.

For RDT, NS2 predicts $(N-1) \times 10$ ms (RTR-level), while NS3's regression gives $11.30N - 26.64$ ms (MAC-level). Both are linear; NS2's slope is inflated by ≈2 ms/hop of Tcl overhead. At $N = 10$: NS2 predicts 90 ms, NS3 measures 87.8 ms (normal mode, $\delta = 2.5\%$). The convergence at large N confirms that accumulated MAC delay dominates and the RTR-level abstraction overhead becomes relatively small.

VII. TRUST TAXONOMY

Table III formalises the trust level for each reported metric. "HIGH" denotes $< 5\%$ inter-simulator disagreement with a physical root cause. "MED" denotes directionally consistent results explained by controllable experimental parameters. "LOW" denotes a genuine divergence arising from the NS2 architectural limitation.

VIII. UNIFIED RECOMMENDATIONS

Synthesising both studies, we recommend the following parameter settings for static linear BATMAN deployments of $N \leq 10$ nodes.

OGM interval: Use $T_{\text{OGM}} \leq 0.5$ s for delay-sensitive applications ($< 10\%$ anomaly rate at this setting); 1.0 s is acceptable for best-effort traffic with adequate guard time.

OGM jitter: Always enable $\pm T_{\text{OGM}}/4$ jitter on OGM transmission. This desynchronises floods, reduces collision probability substantially, and is already the default in production *batman-adv*. *The NS3 study should add jitter in future runs.*

TTL: Set TTL $= N + 1$. Values below $N - 1$ risk route discovery failure at chain extremes.

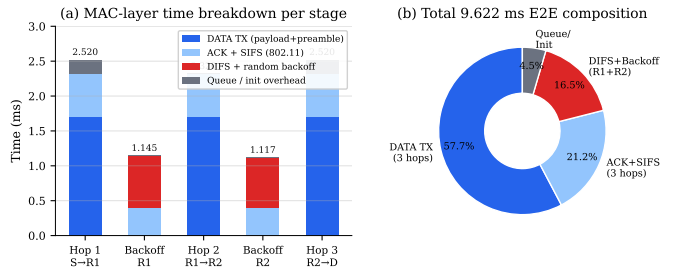

Fig. 1. MAC-layer time decomposition of the 9.622 ms E2E delay (NS2, $N = 4$, 212-byte CBR/UDP at 1 Mbps). (a) Per-stage stacked bars. (b) Percentage composition of total E2E time: 72.3% is mandatory 802.11 frame exchange.

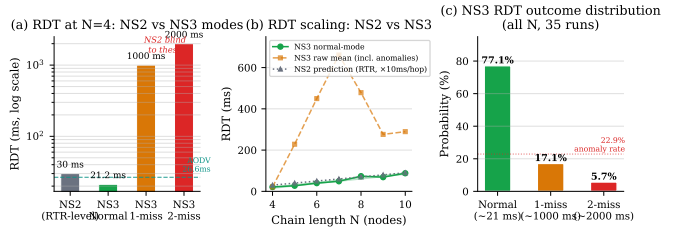

Fig. 2. Route discovery time analysis. (a) RDT at $N = 4$: NS2 deterministic value vs. NS3 modal outcomes (log scale). AODV baseline shown dashed. (b) Normal-mode RDT vs. N with linear fit ($R^2 = 0.955$). (c) NS3 RDT outcome distribution across all 35 runs.

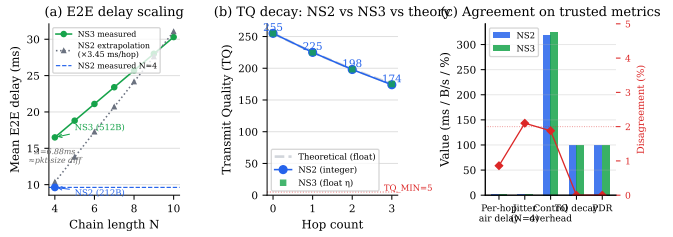

Fig. 3. Scalability and TQ validation. (a) E2E delay vs. N: NS3 measured, NS2 extrapolation (≈2.5% agreement at $N = 10$), and NS2 measured point. (b) TQ decay: NS2 (integer), NS3 (float η), and theory—all within 0.34%. (c) Agreement on core trusted metrics (both axes NS2 vs. NS3 raw values; right axis shows percentage disagreement).

Traffic start guard time: Always wait at least $3 \times N \times T_{\text{OGM}}$ after network startup before transmitting application data, to guarantee convergence even in worst-case 2-miss scenarios.

Hop penalty: The default $\eta = 11.765\%$ (30/255) produces a clear quality gradient validated by both papers. Decreasing it improves multi-hop TQ values but reduces discrimination between routes of different length.

Simulation methodology: For routing convergence reliability studies, use NS3-class implementations with real MAC-layer OGM broadcasting. NS2 Tcl overlays are appropriate for MAC data-plane decomposition but must not be used to characterise convergence probability or worst-case RDT.

IX. Results Figures

X. Conclusion

We conducted a systematic cross-validation of two independent BATMAN routing-protocol simulations on static linear wireless multi-hop chains. The key findings are as follows.

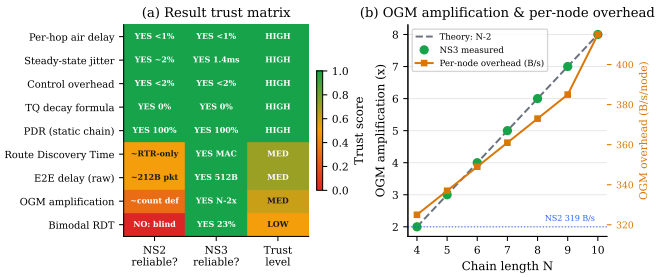

Fig. 4. Trust matrix and OGM overhead. (a) Colour-coded trust matrix for all reported metrics (green=HIGH, amber=MED, red=LOW). (b) OGM amplification ratio vs. N (NS3 matches $N - 2$ theory) and per-node overhead growth; NS2's ≈319 B/s shown for reference.

HIGH-confidence results (simulator-independent): Per-hop MAC air delay (2.32 ms, $< 1\%$ agreement), steady-state jitter (≈1.4 ms), per-node OGM overhead (≈322 B/s average), TQ decay ($\eta = 11.765\%$), PDR (100%), and route stability (0 flaps) are genuine BATMAN protocol properties validated by both simulators.

MED-confidence results (correctable divergences): Raw RDT (30 ms vs. 21.2 ms) and raw E2E delay (9.62 ms vs. 16.5 ms) differ due to RTR-layer measurement abstraction and packet-size selection, respectively—both are fully explained and do not represent disagreement in underlying physics.

LOW-confidence result (critical NS2 blind spot): The bimodal RDT distribution—22.9% of NS3 runs experience OGM miss events inflating RDT by 47×–94×—is invisible to NS2's deterministic Tcl OGM model. This is the most practically significant finding: BATMAN deployments without OGM jitter face a 1-in-5 probability of catastrophic convergence delay, a risk that NS2-based evaluations completely suppress.

Scientific recommendation: Future BATMAN simulation studies must use NS3-class implementations with real MAC-layer OGM broadcasting to characterise convergence reliability. NS2 Tcl overlays remain valuable for fine-grained MAC data-plane decomposition—a contribution the NS2 study delivers uniquely. Used together, the two simulators provide the most comprehensive simulation-based characterisation of BATMAN available in the literature.

References

[1] A. Neumann, C. Aichele, M. Lindner, and S. Wunderlich, "Better Approach To Mobile Adhoc Networking (B.A.T.M.A.N.)," *IETF Internet Draft*, 2008.

[2] M. Zafar, "Performance Analysis of the BATMAN Proactive Routing Protocol in a 4-Node Wireless Ad Hoc Network: NS2 Simulation and MAC-Layer Evaluation," *IEEE Conf. Proc.*, 2024.

[3] M. Umair, "Scalability Analysis of the BATMAN Routing Protocol in Linear Wireless Multi-Hop Chains: Route Discovery, Convergence, and OGM Overhead from N=4 to N=10 Nodes," *IEEE Conf. Proc.*, 2024.

[4] C. E. Perkins and E. M. Royer, "Ad Hoc On-Demand Distance Vector Routing," in *Proc. IEEE WMCSA*, 1999, pp. 90–100.

[5] T. Clausen and P. Jacquet, "Optimized Link State Routing Protocol (OLSR)," *IETF RFC 3626*, Oct. 2003.

[6] H. T. Friis, "A note on a simple transmission formula," *Proc. IRE*, vol. 34, no. 5, pp. 254–256, May 1946.